



Operating Systems

- Course Overview -

Dept. of Computer Science

Hanyang University





- Instructor
 - Prof. Sooyong Kang
 - ITBT 613
 - 2220-1725
 - sykang@hanyang.ac.kr
- TAs
 - Seokhwan Lee (이석환)
 - Classes: 11812, 12913
 - 산학기술관 502
 - 2220-4439
 - huan99@hanyang.ac.kr
 - Junmo Seong (성준모)
 - Classes: 12912
 - ITBT 204-2
 - 2220-4438
 - wnsah814@hanyang.ac.kr



Class Operation: SMART-F

- Theory Class
 - Traditional OS theories
 - We deal with the “concept” of each technique not the “implementation”
 - So, most program code presented in the lecture is not real-world code
 - On-line contents (no offline classes)
 - You must watch weekly online contents before attending hybrid classes!!!
- Hybrid (Theory + Lab.) Class
 - Two hours (exactly, 100 minutes) of offline classes
 - 11812: Thursday (17:00 – 18:40), ITBT 508
 - 12912: Friday (13:00 – 14:40), ITBT 202
 - 12913: Friday (15:00 – 16:40), ITBT 202
 - Short review of the online theory contents + Q&A
 - Practice of key OS theories on the ‘xv6’ educational OS
 - 4 assignments (one easy, two moderate, and one difficult xv6 programming)
 - Detailed introduction will be given by TAs
- Course Materials
 - Weekly online contents (together with PPT slides file) will be uploaded right after the Lab class of the previous week



- Midterm
 - 30%
 - 4월 25일(토) 오전 10시 – 11시 40분 (장소 추후 공지)
 - 시험 범위: 직전 수업 내용까지
- Final
 - 30%
 - 6월 20일(토) 오전 10시 – 11시 40분 (장소 추후 공지)
 - 시험 범위: 전체
- Homework
 - 20% (If three or more assignments have been submitted)
 - F (otherwise, regardless of the exam. scores)
- Homework Quiz
 - 20%
 - A few 'code-level' questions about each homework
 - Will be taken at the first hybrid class after each homework deadline



About Using 'AI-based Code Generators' for Homework

- Ask to yourself !
 - Who do you want to be? (as a CS student)
 - An advanced AI user or an AI developer?
 - If you rely on AI for all the homework, what are you supposed to be?
 - Who are your competetors after graduation? (as a software engineer)
 - Colleague software engineers and AI
 - If you have relied on AI for all the homework, can you prove yourself to be unreplaceable by AI to your recruiters or seniors?
- For what do you study 'Systems'?
 - Core 'Systems' courses: SP, OS, Archi, DB, Network
 - Pursue to '*optimize computer systems*' to expedite application execution
 - Mostly a hardware-coupled 'msec' war, not an 'order of (Big-O)' war in algorithms
 - Systems optimization starts from 'fully understanding' the complicated dynamism of running systems code, which is not covered in classes
 - *Systems code is a mess!* : The effect of a small change in one system component *unpredictably* propagates to many other components → hard to cover in classes
 - The best way to understand such dynamism is to write and verify the effect of systems code by yourself, growing yourself as a '*system-aware software engineer*' that is harder to be replaced by AI than other software engineer

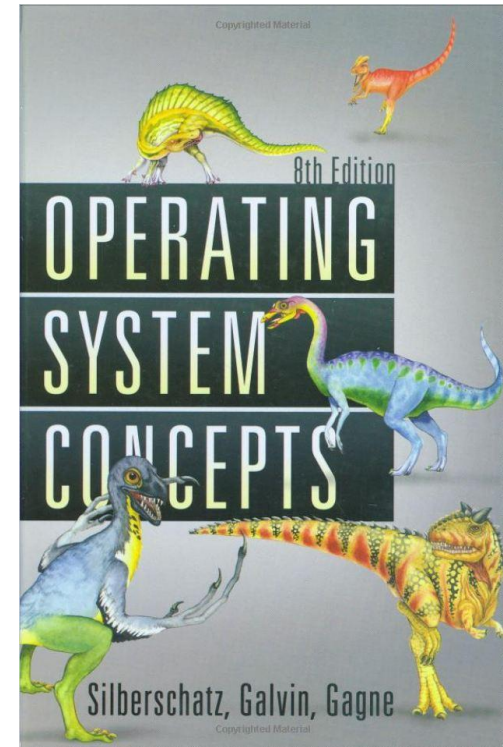
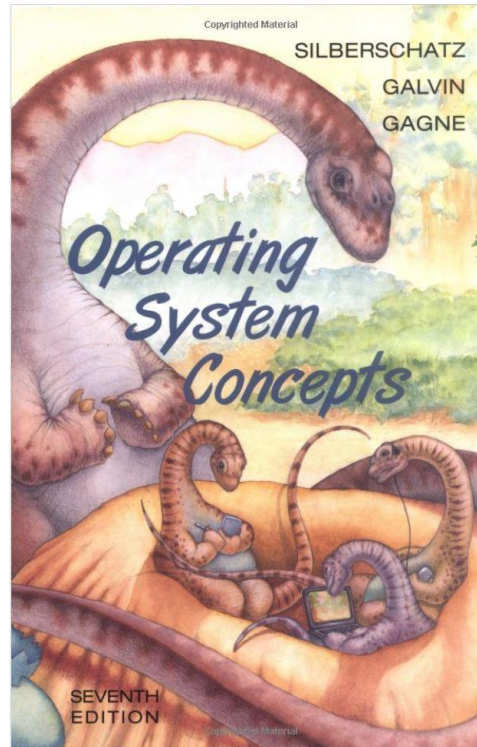
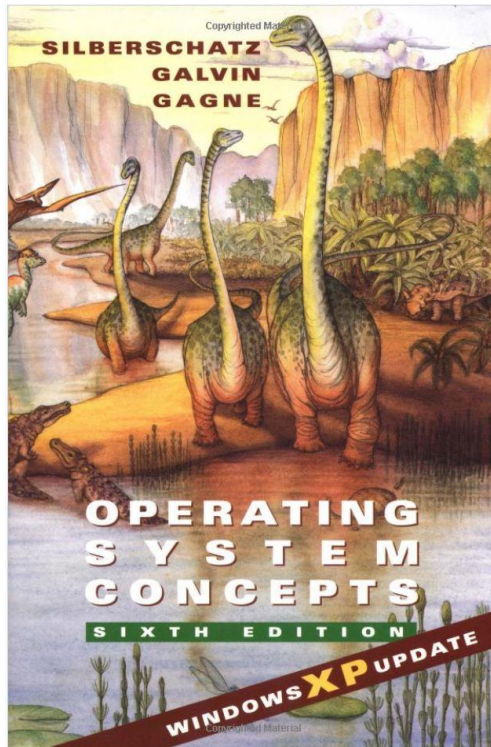


Absence Policy

- Absence policy
 - Three absences will downgrade the final grade by 1 step
 - Ex. Tentative grade (by the original grading policy) A+ → Final grade A
 - More than 8 absences w/o due notice: definitely F
 - Watch weekly online contents until the day before the lab class (23:59 firm deadline)!!!
 - Otherwise, will be regarded as an absence to the theory class
- Late policy (applied only in Lab. classes)
 - Definition of the 'Late': Appear within 15 min. after the class time
 - After 15 minutes, no late but absence!
 - Three times of late will be regarded as an absence
 - Binary policy (attend or absence) for theory classes
- Escape policy: rigorous discipline for dishonesty (in Lab. classes)
 - Definition of the 'Escape': disappear w/o due notice after the attendance check and before class ends
 - An escape will be regarded as two absences
 - More than one escape: definitely F

Textbook

- Operating System Concepts, Any (>5th) ed., Silberschatz, Galvin and Gagne, John Wiley & Sons, Inc



Operating System: An Intuitive Concept



Operating System: An Intuitive Concept

- What can you do for your new smart TV against your nephew?



**You are doing exactly what OS does
for computer HW against users !**



What You Do and What OS Does

- You enables your nephews to use TV
 - Conveniently
 - So that even your little nephew can use smart TV
 - Efficiently
 - Arbitrate between two nephews combating for channels
 - Effectively
 - Prevent from meaningless button pressing
 - Securely
 - Keep TV secure against rash behaviors of nephew
 - Protectively
 - Keep your nephew out of adults channels

- OS enables users to use computer
 - Conveniently
 - So that even fresh users can use the computer
 - Efficiently
 - Arbitrate between multiple jobs combating for resources
 - Effectively
 - Prevent from meaningless resource usage
 - Securely
 - Keep HW secure against malfunctioning jobs
 - Protectively
 - Keep other users out of your files



OS Strategy to achieve the goals

Cheating

which is

- Solid
- Sustainable
- Fault-tolerant
- and unpunishable



*To make students truly understand that
OS is a collection of cheating schemes*



- Process management: *Multitasking*
 - Cheating by fast and repetitive task switches
- Memory management: *Virtual Memory*
 - Cheating by a phantom space provision
- File and Storage management: *Fast I/O*
 - Cheating by buffering and caching



Conclusion and Q&A

Understood?

Then you are ready to take this Operating System course!

Good Luck to you!